

Run inputs

scientific request +
deployment context

experiment config

method + task + seed

runtime metadata

device + partition

The control plane decides placement
and passes only the runtime
metadata needed by the app.

fedctl_research Flower app

shared task code with method-family hooks

Server-app

run-config resolver

loads method, task, budgets,
evaluation hooks

global model + task

constructs task interface and
centralized evaluator

Client-app

node resolver

reads node config, env metadata,
partition id

local train / eval

loads partition, applies
model rate, reports metrics

shared task API: models, partitions, local train/evaluate

method hooks

full-model sync

model-heterogeneous

buffered async

Outputs

method-layer evidence for
evaluation

W&B summaries

global curves, final scores,
target state

Structured artifacts

event-level rows and per-client
diagnostics

Device-labelled metrics

training time, model rate, update
and weight share